

CHEMISTRY 30

Advanced Diploma Preparation

UNIT D — CHEMICAL EQUILIBRIUM

Study Guide | Version 2 Advanced Synthesis Set

 Course: Chemistry 30  Focus: Quadratic ICE, buffers, titrations, mixed equilibria  Level: Upper diploma difficulty	Name: _____ Class: _____ Date: _____
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1 Unit / Module Objectives

Topic Area	Diploma-Level Objective
Quadratic ICE Systems	Solve equilibrium concentrations requiring quadratic solutions when small-x approximations are not justified.
Reaction Quotient Analysis	Compare Q and K to predict spontaneous direction of shift before equilibrium is re-established.
Temperature Dependence	Predict how temperature changes affect equilibrium position and the value of K.
Ka and Kb Derivation	Determine Ka or Kb from equilibrium concentration data for weak acids and weak bases.
Henderson–Hasselbalch Equation	Apply Henderson–Hasselbalch reasoning to calculate buffer pH and target base-to-acid ratios.
Buffer Disturbance	Calculate buffer pH after strong-acid or strong-base addition using stoichiometric adjustment followed by equilibrium logic.
Polyprotic Systems	Interpret diprotic titration curves, equivalence points, half-equivalence points, and dominant species.
Mixed Equilibria	Analyze coupled equilibria involving Ka, Kb, conjugate pairs, and Kw.

Graph Interpretation	Infer applied stresses from concentration-time graphs and titration data.
Diploma Strategy	Validate final answers using sig figs, limiting assumptions, and chemical reasonableness.

2 Student Self-Verification Checklist

Topic	Self-Verification Statement
Quadratic ICE Tables	☐ I can solve an equilibrium problem with the quadratic formula when the 5% rule fails.
Q vs K Prediction	☐ I can compare Q and K before doing algebra and predict the correct shift direction.
Temperature Effects	☐ I can justify whether K increases or decreases when temperature changes.
Ka Calculations	☐ I can calculate Ka from equilibrium concentrations and classify relative acid strength.
Henderson–Hasselbalch	☐ I can calculate buffer pH and rearrange the equation to obtain a required ratio.
Buffer Disturbance	☐ I can adjust moles first after adding strong acid/base and then recalculate pH.
Polyprotic Systems	☐ I can interpret a diprotic acid titration curve and identify both equivalence points.
Weak Bases	☐ I can calculate pOH and then convert correctly to pH for a weak base.
Graph Analysis	☐ I can identify the stress shown by an instantaneous jump followed by a gradual re-equilibration.
Common Misconceptions	☐ I can recognize when a weak acid/base should not be treated as a complete reaction.

3 Diploma-Level Practice Questions with Fully Worked Solutions

Advanced Set Focus

This version emphasizes quadratic ICE solutions, Henderson–Hasselbalch reasoning, stoichiometry-to-equilibrium crossover, diprotic titration interpretation, and mixed-equilibrium logic.

Question	1. A 0.500 mol/L solution of HA has $K_a = 4.0 \times 10^{-5}$. Calculate the equilibrium $[H_3O^+]$ without using a small-x approximation.
Step 1 — Identification	This is a weak-acid equilibrium problem with a relatively concentrated acid, but the instructions explicitly require the non-approximate method.
Step 2 — Principle	Use an ICE table and the acid-ionization expression $K_a = [H_3O^+][A^-]/[HA]$.
Step 3 — Explicit Reasoning	Let $x = [H_3O^+]$ at equilibrium. $HA \rightleftharpoons H_3O^+ + A^-$ Initial: 0.500, 0, 0 Change: $-x$, $+x$, $+x$ Equilibrium: $0.500 - x$, x, x $K_a = x^2/(0.500 - x) = 4.0 \times 10^{-5}$ $x^2 = 2.0 \times 10^{-5} - 4.0 \times 10^{-5}x$ $x^2 + 4.0 \times 10^{-5}x - 2.0 \times 10^{-5} = 0$ Using the quadratic formula gives $x = 4.45 \times 10^{-3}$ mol/L (negative root rejected).
Step 4 — Validation	The value is much smaller than 0.500 mol/L, so the result is chemically reasonable for a weak acid. The exact method confirms the concentration without relying on approximation.
Step 5 — Final Answer	$[H_3O^+] = 4.45 \times 10^{-3}$ mol/L
Common Error	Using $x = \sqrt{K_a C}$ automatically when the question explicitly requires the exact solution.
Exam Strategy	When the exam flags a non-approximate method, write the quadratic cleanly and reject the negative root only after solving.

Question	2. For a reaction mixture, $Q = 1.2 \times 10^{-3}$ and $K = 5.0 \times 10^{-2}$. Predict the direction of shift and explain why no ICE table is needed yet.
Step 1 — Identification	This is a reaction-quotient comparison problem, not a concentration-calculation problem.
Step 2 — Principle	If $Q < K$, the system must produce more products to reach equilibrium. If $Q > K$, it must produce more reactants.
Step 3 — Explicit Reasoning	Compare the two values directly: $1.2 \times 10^{-3} < 5.0 \times 10^{-2}$

	Therefore the mixture has too few products relative to the equilibrium ratio. The forward reaction is favoured until Q rises to K.
Step 4 — Validation	No ICE table is needed because the question asks only for direction, not for equilibrium concentrations.
Step 5 — Final Answer	The system shifts right, toward products.
Common Error	Starting algebra before first comparing Q and K.
Exam Strategy	Always do the Q-versus-K comparison first. It often gives the whole answer in one line.

Question	3. An exothermic equilibrium has $\Delta H < 0$. Predict the effect of increasing temperature on the equilibrium position and on K.
Step 1 — Identification	This is a qualitative equilibrium-stress problem involving temperature.
Step 2 — Principle	For an exothermic forward reaction, heat behaves like a product. Increasing temperature adds product-side stress.
Step 3 — Explicit Reasoning	Adding heat causes the system to shift left to consume some of that added heat. Because equilibrium composition changes with temperature, the equilibrium constant also changes. For an exothermic system, higher temperature decreases product favourability, so K becomes smaller.
Step 4 — Validation	Catalysts do not change K, but temperature does. That distinction is the key validation point.
Step 5 — Final Answer	Increasing temperature shifts the equilibrium left and decreases K.
Common Error	Saying that K never changes. Only concentration and pressure changes leave K unchanged; temperature does not.
Exam Strategy	Write heat on the appropriate side mentally. Then decide which direction reduces the temperature stress.

Question	4. A buffer contains 0.40 mol/L HA and 0.25 mol/L A^- . If $K_a = 1.8 \times 10^{-5}$, calculate the buffer pH.
Step 1 — Identification	This is a weak-acid/conjugate-base buffer because both HA and A^- are present in significant amounts.
Step 2 — Principle	Use Henderson–Hasselbalch: $pH = pK_a + \log([A^-]/[HA])$.
Step 3 — Explicit Reasoning	$pK_a = -\log(1.8 \times 10^{-5}) = 4.74$ $pH = 4.74 + \log(0.25/0.40)$

	$\text{pH} = 4.74 + \log(0.625)$ $\text{pH} = 4.74 - 0.204 = 4.54$
Step 4 — Validation	Because $[\text{A}^-]$ is smaller than $[\text{HA}]$, the pH should be slightly below pK_a . The answer 4.54 matches that expectation.
Step 5 — Final Answer	pH = 4.54
Common Error	Reversing the ratio and using $[\text{HA}]/[\text{A}^-]$.
Exam Strategy	In Henderson–Hasselbalch, it is always base over acid.

Question	5. What ratio $[\text{A}^-]/[\text{HA}]$ is required to prepare a buffer at pH 6.00 if $\text{pK}_a = 5.20$?
Step 1 — Identification	This is a rearranged Henderson–Hasselbalch problem.
Step 2 — Principle	Use $\text{pH} = \text{pK}_a + \log([\text{A}^-]/[\text{HA}])$ and solve for the ratio.
Step 3 — Explicit Reasoning	$6.00 = 5.20 + \log([\text{A}^-]/[\text{HA}])$ $0.80 = \log([\text{A}^-]/[\text{HA}])$ $[\text{A}^-]/[\text{HA}] = 10^{0.80} = 6.31$
Step 4 — Validation	Because the target pH is above pK_a , the base form should predominate. A ratio greater than 1 is therefore reasonable.
Step 5 — Final Answer	$[\text{A}^-]/[\text{HA}] = 6.31$
Common Error	Stopping at 0.80 and forgetting to remove the logarithm.
Exam Strategy	After isolating the log term, always exponentiate to obtain the actual ratio.

Question	6. A 1.00 L buffer initially contains 0.30 mol HA and 0.25 mol A^- . Then 0.010 mol HCl is added. Calculate the new pH if $K_a = 1.8 \times 10^{-5}$.
Step 1 — Identification	This is a buffer-disturbance crossover problem. Strong acid reacts first by stoichiometry; buffer pH is calculated second.
Step 2 — Principle	H^+ from HCl reacts completely with A^- to form HA. After the stoichiometric change, use Henderson–Hasselbalch.
Step 3 — Explicit Reasoning	Initial moles: $\text{HA} = 0.30$, $\text{A}^- = 0.25$ Reaction with 0.010 mol HCl: $\text{A}^- \rightarrow 0.25 - 0.010 = 0.24 \text{ mol}$ $\text{HA} \rightarrow 0.30 + 0.010 = 0.31 \text{ mol}$ Since the total volume is still 1.00 L, concentrations are 0.24 mol/L and 0.31 mol/L. $\text{pK}_a = 4.74$ $\text{pH} = 4.74 + \log(0.24/0.31)$

	$\text{pH} = 4.74 + \log(0.774)$ $\text{pH} = 4.74 - 0.111 = 4.63$
Step 4 — Validation	The pH should decrease slightly because strong acid was added, but not dramatically because the solution is buffered. A change from near pKa to 4.63 is consistent.
Step 5 — Final Answer	New pH = 4.63
Common Error	Using the original buffer concentrations without first adjusting the moles for the added HCl.
Exam Strategy	For any buffer disturbance, do stoichiometry first and equilibrium second.

Question	7. A diprotic acid titration curve shows two distinct equivalence points. Explain what each equivalence point represents and what can be determined from the half-equivalence points.
Step 1 — Identification	This is a titration-curve interpretation problem for a polyprotic acid.
Step 2 — Principle	Each equivalence point corresponds to the neutralization of one acidic proton. Half-equivalence points reveal pKa values because [acid form] = [conjugate base form] at those locations.
Step 3 — Explicit Reasoning	For H_2A : • First equivalence point: one proton per molecule has been neutralized, so HA^- predominates. • Second equivalence point: both acidic protons have been neutralized, so A^{2-} predominates. At the midpoint before the first equivalence point, $\text{pH} = \text{pKa}_1$. At the midpoint between the first and second equivalence points, $\text{pH} = \text{pKa}_2$.
Step 4 — Validation	Two buffering regions and two equivalence points are exactly what should appear for a diprotic acid whose dissociation steps are sufficiently separated.
Step 5 — Final Answer	The first equivalence point corresponds to $\text{H}_2\text{A} \rightarrow \text{HA}^-$, the second to $\text{HA}^- \rightarrow \text{A}^{2-}$, and the half-equivalence points provide pKa_1 and pKa_2.
Common Error	Assuming every acid titration has only one equivalence point.
Exam Strategy	Count the ionizable protons. That tells you how many equivalence points are possible.

Question	8. Calculate the pH of 0.10 mol/L $\text{NH}_3(\text{aq})$ if $K_b = 1.8 \times 10^{-5}$.
Step 1 — Identification	This is a weak-base equilibrium problem requiring pOH first, then pH.
Step 2 — Principle	Use $K_b = [\text{NH}_4^+][\text{OH}^-]/[\text{NH}_3]$ with an ICE table.
Step 3 — Explicit Reasoning	Let $x = [\text{OH}^-]$ $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ Initial: 0.10, 0, 0 Change: $-x$, $+x$, $+x$ Equilibrium: $0.10 - x$, x, x $K_b = x^2/(0.10 - x) = 1.8 \times 10^{-5}$ Because x is small relative to 0.10, $x \approx \sqrt{(1.8 \times 10^{-5})} = 1.34 \times 10^{-3}$ mol/L $\text{pOH} = -\log(1.34 \times 10^{-3}) = 2.87$ $\text{pH} = 14.00 - 2.87 = 11.13$
Step 4 — Validation	The solution should be basic and well above pH 7. A pH near 11 is consistent with a weak base of moderate concentration.
Step 5 — Final Answer	pH = 11.13
Common Error	Stopping at pOH or subtracting in the wrong direction.
Exam Strategy	For weak bases, circle pOH as an intermediate answer so you remember to convert to pH.

Question	9. A concentration-time graph shows that the product concentration increases instantaneously, then gradually decreases to a new plateau. Identify the most likely stress.
Step 1 — Identification	This is a graph-based equilibrium-diagnosis problem.
Step 2 — Principle	An instantaneous change signals that a species was directly added or removed. A gradual change reflects the equilibrium shift that follows.
Step 3 — Explicit Reasoning	Because the product concentration jumps upward immediately, the most likely direct stress is that product was added. The gradual decrease afterward shows the system shifting left to consume part of the excess product and re-establish equilibrium.
Step 4 — Validation	A temperature or catalyst change would not produce that same immediate one-species jump pattern.
Step 5 — Final Answer	The most likely stress is the addition of product.
Common Error	Describing only the later shift and ignoring the meaning of the instantaneous jump.
Exam Strategy	Separate the graph into two events: the immediate disturbance and the gradual equilibrium response.

Question	10. During titration of H_2A with strong base, what species is dominant at the second equivalence point?
Step 1 — Identification	This is a species-dominance question in a diprotic titration.
Step 2 — Principle	At each equivalence point, one stoichiometric proton-removal stage is complete.
Step 3 — Explicit Reasoning	By the second equivalence point, the first proton and the second proton have both been neutralized. Therefore the fully deprotonated form A^{2-} is the dominant species in solution, aside from spectator ions and water.
Step 4 — Validation	This matches the stoichiometric sequence $\text{H}_2\text{A} \rightarrow \text{HA}^- \rightarrow \text{A}^{2-}$.
Step 5 — Final Answer	A^{2-} is the dominant acid-derived species at the second equivalence point.
Common Error	Choosing HA^- because it appears at the first equivalence point and forgetting that a second full neutralization has occurred.
Exam Strategy	Track the proton count removed at each stage rather than memorizing isolated curve facts.